Credit Risk Portfolio Analytics

Solution overview | Case study

Objective

Build an interactive view of credit portfolio performance and provide practical tools for identifying accounts that may warrant further review. The solution combines descriptive analysis, behavioral classification, anomaly detection, and a grounded question-answering interface.

Portfolio summary

Loan accounts	14,685	Unique customers	10,000
Observed default rate	16.10%	Original principal	$601.12M

Findings from the portfolio

- Mortgage accounts have the highest observed default rate (24.65%) and account for approximately 52.3% of original principal.
- The unemployed segment has a 34.96% default rate. This is an observed association, not evidence of causation.
- Default rates decrease across income quintiles, from 25.49% in the lowest-income group to 9.50% in the highest.
- Accounts aged 25 or younger show a 22.16% default rate. The 65+ segment is small (91 accounts), so its estimate is less stable.

Model results

Model	ROC-AUC	KS	Gini	Recall at 0.50
Logistic Regression	0.6768	0.2703	0.3537	0.610
XGBoost	0.6500	0.2306	0.2999	0.488

The Logistic Regression model was selected for the prototype because it performed better on ROC-AUC, KS, Gini, and recall in the evaluated split. Performance is modest; a review threshold should be chosen using validation results and business costs rather than treated as a fixed lending rule.

Dashboard capabilities

- Portfolio filters and segment charts.
- Account-level risk score with SHAP contribution view.
- Isolation Forest anomaly flags, review statuses, and downloadable review queue.
- Gemini-powered Q&A constrained to calculated portfolio results; unsupported questions are declined.

Data quality and limitations

Missing demographic/employment values and inconsistent category labels were identified during validation. Collections events before loan origination were excluded from linked features. The dataset lacks a reliable default date and bureau as-of date, so temporal leakage cannot be fully ruled out. These results are retrospective behavioral classification—not validated origination underwriting. Production use would require temporal validation, calibration, fairness review, governance, and monitoring.